

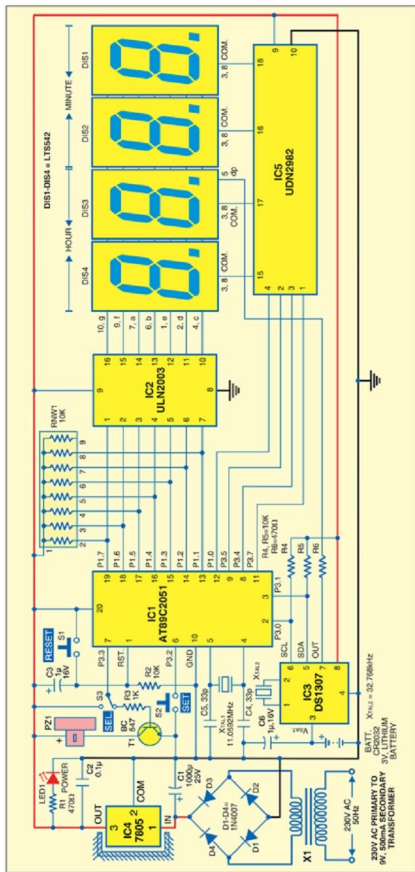


Fig. 2: Pin configurations of 7805 and BC547

connected at its pin 3 along with 1µF capacitor for battery backup. Pins 5 and 6 are pulled up to 5V by resistors R5 and R4 and connected to pins P3.1 and 3.0 of the microcontroller, respectively, for serial communication between the RTC and the microcontroller.

To derive the supply power for the circuit, the 230V, 50Hz AC mains is stepped down by transformer X1 to deliver a secondary output of 9V, 500mA. The transformer's output is rectified by a full-wave bridge rectifier comprising diodes D1 through D4, filtered by capacitor C1 and regulated by IC 7805 (IC4). Capacitor C2 bypasses the ripples present in the regulated output. LED1 acts as the power indicator. Resistor R1 acts as the current-limiter for LED1.

An actual-size, single-side PCB for the microcontroller-based clock using DS1307 is shown in Fig. 3 and its component layout in Fig. 4.

## Setting of current time and alarm time

To set the alarm time, press switch S2 connected to pin P3.2 of the microcontroller and keep it pressed until the display changes and reaches the required alarm time, say, 7:30. Now connect the pole of SPDT switch S3 to Vcc momentarily (which makes pin P3.3 of the microcontroller high) and bring it back to the initial position. Immediately, the alarm sounds for a minute. The alarm time is thus set at 7:30.

To set the current time, keep switch S2 pressed until the time display reaches the current time. Release it when the current time is set.

Once the time is set and the clock is working, there is no need to set the time again even after switching off the circuit. This is because the RTC works off the 3V battery connected to its pin 3.

Fig. 1: Circuit of microcontroller-based clock using DS1307